

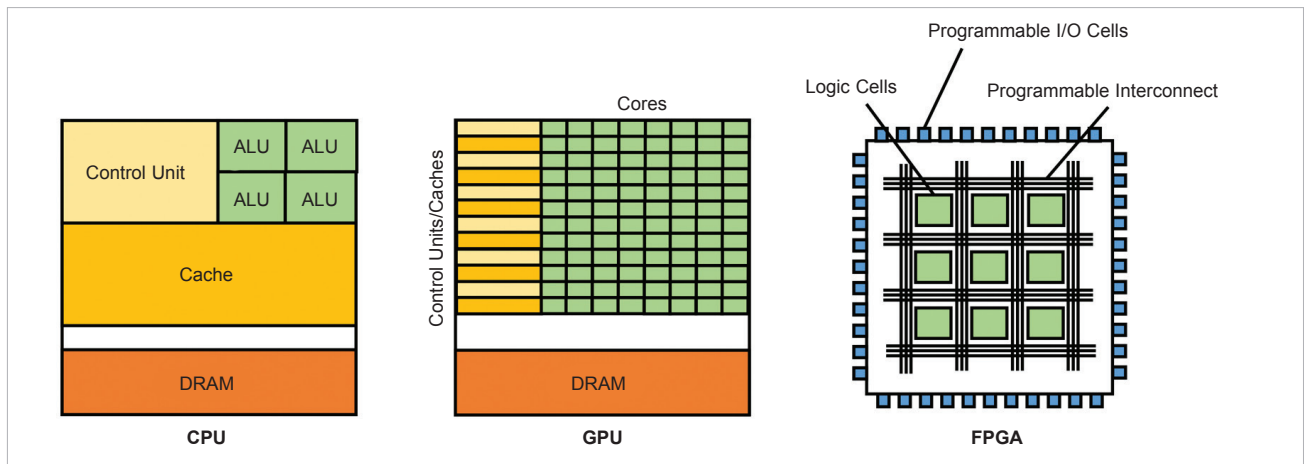


Fig. 2: Different types of processors

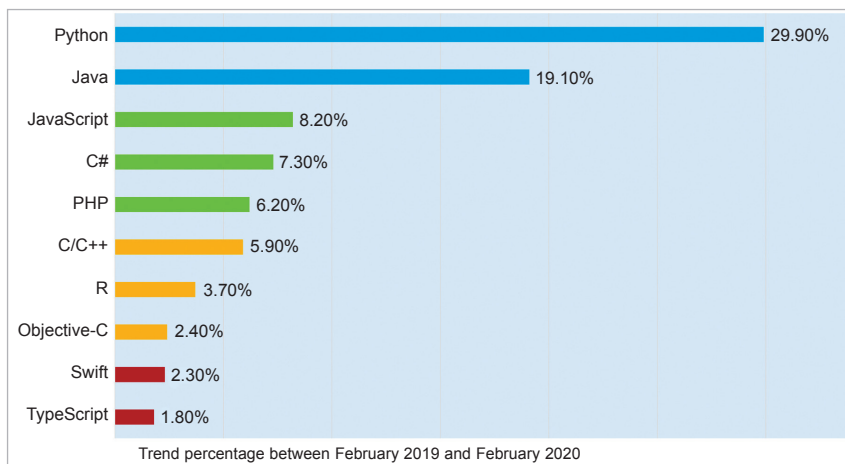


Fig. 3: Trend percentages of programming languages (Reference: www.statista.com)

grammed as per the purpose, and its architecture can be customised for the specific agent in ML. An FPGA chip can contain thousands of memory units, which are more than in a GPU, and can give better throughputs. Another advantage of using an FPGA for ML is hardware acceleration, which can accelerate certain parts of an algorithm, making it more efficient than a GPU.

In order to take full advantage of the hardware, good ML software can help build ML models as per the requirement. There is a lot of such software available online—both proprietary and free. Among the free software, TensorFlow, Shogun, Apache Mahout, PyTorch, KNIME, and Keras are more widely used and are the popular ones.

- TensorFlow can help build ML solutions through its extensive interface of a CUDA GPU. It provides support and functions for various applica-

tions of ML such as computer vision, NLP, and reinforcement learning. This software is best suited for beginners in ML, and is also used for education purposes.

- Shogun is free software that supports languages like Python, R, Scala, C#, Ruby, etc. It supports vector agents, dimensionality reduction, clustering algorithms, hidden Markov models, and linear discriminant analysis.

- Apache Mahout is popular software that provides expressive Scala DSL and a distributed linear algebra framework for deep learning computations and native solvers for CPUs, GPUs as well as CUDA accelerators.

- PyTorch was developed by Facebook's AI Research (FAIR) lab and is mainly used for ML applications such as computer vision and natural language processing. It provides Tensor computing (like NumPy) with strong acceleration via GPUs, and supports

deep neural networks built on a tape based automatic differentiation system. The Tesla Autopilot (advanced driver-assistance system) used in Tesla cars was built using PyTorch.

- KNIME or the Konstanz Information Miner is free software that can do data analysis and reporting using ML and data mining. It integrates various components for agent learning and data mining through its modular data pipelining 'Lego of Analytics' concept. It provides a graphical user interface (GUI) and Java database connectivity (JDBC) features for blending various data sources for modelling, data analysis, and visualisation without or with only minimal programming. It has been used in areas like pharmaceutical research, CRM customer data analysis, business intelligence, text mining, and financial data analysis.

- Keras provides a Python interface for artificial neural networks and is well known for its modularity, speed, and ease of use. It supports backends like TensorFlow, Microsoft Cognitive Toolkit, Theano, and PlaidML. It is designed to enable fast experimentation with deep neural networks and is user friendly too.

Python for ML

Many types of programming languages like Python, C/C++, Java/JavaScript and R are used for ML but Python is the most widely used because of its simplicity and features. It was created in the late 1980s and was first released in 1991, by Guido van Rossum, as a successor to the ABC programming language.